

Dynamic Guideline: WHO-Based Cardiovascular Disease - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope note

For this report, “cardiovascular disease” includes WHO guidance on cardiovascular risk prevention and management, hypertension, rheumatic fever/rheumatic heart disease, HEARTS primary-care implementation, and regional CVD monitoring. The latest **WHO clinical guideline** identified in the supplied evidence base is the **WHO guideline on the prevention and diagnosis of rheumatic fever and rheumatic heart disease**, published **29 October 2024** [WHO RF/RHD guideline](#). A newer WHO CVD-related implementation/monitoring document, **SEAHEARTS 2030: Monitoring Framework**, is dated **21 January 2026**, but it is a monitoring framework rather than a clinical treatment guideline [WHO cardiovascular diseases topic page](#).

The WHO publication titled “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” is a **2025 WHO guideline-development case study** about the 2022 abortion care guideline, not a cardiovascular guideline [WHO IRIS](#). It is included here only because it is relevant to WHO guideline methodology and stakeholder engagement, including future CVD guideline development.

Exactly 5 WHO cardiovascular-related guidelines or guideline-equivalent WHO documents synthesized

No.	WHO document	Publication / latest identified date	Target population	Scope and objectives	Status for this report
1	WHO guideline on the prevention and diagnosis of rheumatic fever and rheumatic heart disease	29 October 2024	Children, adolescents, adults, pregnant women, and populations at moderate/high RHD risk	Primary prevention of RF/RHD, secondary prophylaxis, screening/diagnosis, and RF anti-inflammatory management WHO RF/RHD guideline	Latest WHO CVD clinical guideline

No.	WHO document	Publication / latest identified date	Target population	Scope and objectives	Status for this report
2	Guideline for the pharmacological treatment of hypertension in adults	24 August 2021	Adults with hypertension	Pharmacological treatment of hypertension to reduce CVD risk WHO CVD programme	WHO clinical guideline
3	HEARTS: Technical package for cardiovascular disease management in primary health care — risk-based CVD management	Date not extractable from supplied summary; active WHO tool as of 29 April 2026	Primary-care programmes, especially in resource-limited and LMIC settings	Total cardiovascular-risk assessment and management using country-specific risk charts WHO HEARTS risk-based CVD management	WHO implementation guideline-equivalent
4	Prevention of cardiovascular disease: guidelines for assessment and management of total cardiovascular risk	1 January 2007	People at high CVD risk without prior cardiovascular events	Reducing premature death and disability from coronary heart disease, cerebrovascular disease, and peripheral vascular disease by total-risk assessment and preventive therapy WHO prevention guideline	WHO guideline
5	HEARTS: Technical package for cardiovascular disease management in primary health care — tool for developing a consensus protocol for treatment of hypertension	1 April 2021	Primary-care health systems and programme managers implementing hypertension treatment protocols	Development of standardized drug- and dose-specific hypertension treatment protocols in primary care WHO HEARTS hypertension protocol tool	WHO implementation guideline-equivalent

1.2 Key Recommendations

Interpretation of GRADE ratings in this report

- Where WHO-stated GRADE ratings are available in the supplied summaries, they are reported as such.
- Where the supplied summaries identify a WHO guideline or implementation tool but do **not** provide extractable WHO GRADE certainty/strength, a conservative report-level GRADE assessment is provided and explicitly marked as **“report-assigned”** rather than an official WHO grade.
- Implementation frameworks and monitoring tools are not always designed as GRADE-based clinical guidelines; therefore, recommendations from HEARTS and SEAHEARTS-type materials are graded for implementation certainty rather than drug efficacy.

1.2.1 Latest WHO clinical CVD guideline: rheumatic fever and rheumatic heart disease

Recommendation	GRADE Evidence Quality	Strength
Use the Jones criteria for diagnosis of rheumatic fever in children, adolescents, and adults with suspected RF.	Low	Strong
Where standard echocardiography is unavailable, use handheld echocardiography to diagnose RF-carditis.	Very Low	Strong
Where standard echocardiography is unavailable, use handheld echocardiography to diagnose RHD.	Moderate	Strong
In moderate/high RHD-risk populations or settings, consider echocardiographic screening during antenatal care for pregnant women to improve early detection.	Very Low	Conditional
In moderate/high RHD-prevalence populations, implement standard or handheld echocardiographic screening for early RHD detection among children and adolescents aged 5–19 years .	High	Strong
Children, adolescents, and adults diagnosed with RF or RHD should receive antibiotic prophylaxis to prevent RF recurrence.	Moderate	Strong
Children and adolescents meeting minimum criteria for RHD on echocardiographic screening should receive antibiotic prophylaxis to prevent progression.	Moderate	Strong
No recommendation can be made for or against a specific clinical prediction rule for suspected group A streptococcal skin infection.	Very Low	No recommendation
No recommendation can be made for or against antibiotic treatment of skin and skin-structure infections specifically to prevent RF/RHD.	Very Low	No recommendation

Sources: WHO RF/RHD guideline and NCBI guideline summary [WHO RF/RHD guideline](#), [NCBI Bookshelf](#).

1.2.2 Broader WHO cardiovascular disease prevention and management recommendations

WHO recommendation or guidance area	Guideline/source	GRADE Evidence Quality	Strength
Use absolute total cardiovascular risk to guide preventive decisions and treatment intensity rather than treating isolated risk factors alone.	WHO 2007 total-risk guideline WHO prevention guideline	Low, report-assigned; official GRADE table not extractable from supplied summaries	Strong
Provide lifestyle counselling and risk-factor reduction for tobacco use, unhealthy diet, harmful alcohol use, inadequate physical activity, hypertension, high cholesterol, and high blood glucose.	WHO CVD topic/fact sheet WHO CVD topic , WHO CVD fact sheet	Moderate, report-assigned	Strong
Treat hypertension, diabetes, and high blood lipids when clinically indicated to reduce risk of heart attack and stroke.	WHO CVD fact sheet WHO CVD fact sheet	Moderate, report-assigned	Strong
Strengthen primary-care delivery of CVD prevention and control through the Global HEARTS Initiative and HEARTS technical package.	WHO CVD fact sheet and HEARTS module WHO CVD fact sheet , WHO HEARTS risk-based CVD management	Low, report-assigned implementation evidence	Strong
Use standardized hypertension treatment protocols in primary health care.	WHO HEARTS hypertension protocol tool WHO HEARTS hypertension protocol tool	Low, report-assigned implementation evidence	Strong
Avoid broad population-level CVD screening programmes where WHO instead recommends targeted case-finding among higher-risk primary-care populations.	WHO Health Evidence Network synthesis report summary WHO HEN report	Low, report-assigned	Conditional
Monitor CVD programme progress using defined indicators and milestones, including SEAHEARTS 2030 monitoring indicators where applicable.	SEAHEARTS 2030 Monitoring Framework WHO CVD topic	Very Low, report-assigned implementation certainty	Conditional

1.3 Guideline Development Methodology

WHO RF/RHD guideline methodology

The 2024 WHO RF/RHD guideline was developed according to WHO guideline-development procedures, including the **WHO handbook for guideline development**, and met international standards for evidence-informed guidelines [WHO RF/RHD guideline PDF](#).

Key methodological features included:

1. Scope and key questions

- The guideline addressed:
 - Primary prevention of RF/RHD.
 - Secondary prevention of recurrent RF/RHD.
 - Screening and diagnosis of early RHD.
 - Anti-inflammatory treatment for RF [WHO RF/RHD guideline](#).

2. Evidence review

- WHO used systematic reviews and evidence assessment, especially for areas where evidence certainty was low or very low [WHO RF/RHD guideline PDF](#).

3. Consensus method

- Recommendations were formulated by a guideline development group with diverse expertise and perspectives [WHO RF/RHD guideline PDF](#).

4. Conflict of interest management

- Declarations of interest were assessed and managed according to WHO policy to support transparency and minimize bias [WHO RF/RHD guideline PDF](#).

5. Review and quality assurance

- The process included drafting, review, quality-assurance review, publication, dissemination, and planned evidence monitoring [WHO RF/RHD guideline PDF](#).

Youth and lived-experience engagement

WHO's 2025 publication "**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**" demonstrates how young people can contribute to WHO guideline decision-making through lived experience, fresh perspectives, and innovative solutions [WHO publication page](#). Although this case study is not cardiovascular-specific, its principles are relevant to future CVD guidelines, particularly conditions such as RHD that often begin in childhood and peak diagnostically in young adults aged **20–39 years** [WHO RF/RHD guideline](#).

A broader WHO meaningful-engagement framework also included lived-experience input from people with conditions including diabetes and rheumatic heart disease [PMC/NIH](#).

II. Evidence Summary After WHO Latest Guideline Release (WHO guideline on the prevention and diagnosis of rheumatic fever and rheumatic heart disease) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Reference guideline

The reference guideline for this update is the **WHO guideline on the prevention and diagnosis of rheumatic fever and rheumatic heart disease**, published **29 October 2024** [WHO RF/RHD guideline](#).

Search strategy

The supplied research process searched WHO sources and secondary indexing sources for evidence from **29 October 2024 to 29 April 2026**, including:

- WHO publication pages and WHO IRIS.
- NCBI Bookshelf summaries of WHO guideline recommendations.
- Guideline Central summaries.
- Global Heart.
- Peer-reviewed systematic reviews, meta-analyses, and narrative reviews identified in the supplied results.
- Clinical trial registry signals, where available.

Search concepts included:

- Rheumatic fever.
- Rheumatic heart disease.
- WHO guideline.
- Diagnosis.
- Echocardiography.
- Jones criteria.
- Benzathine penicillin G.
- Antibiotic prophylaxis.
- Implementation in endemic settings.
- Systematic review.
- Randomized controlled trial.
- Clinical trial registry.

Inclusion criteria

Evidence was included if it was:

- Published or identified after **29 October 2024**.
- Relevant to RF/RHD prevention, diagnosis, secondary prophylaxis, screening, safety, or implementation.
- WHO-authored, WHO-referenced, peer-reviewed, or indexed in a reputable evidence source.

Exclusion criteria

Evidence was excluded if it was:

- Not related to CVD/RF/RHD.
- A non-WHO guideline unrelated to WHO recommendations.
- A duplicate summary without new information.

- A source without enough information to assess relevance.

Overall evidence conclusion

No official WHO update, corrigendum, replacement guideline, or formal recommendation change was identified through **29 April 2026**. WHO’s stated plan remains to monitor new evidence and reassess the guideline at **5 years from publication** unless earlier evidence triggers an update [WHO RF/RHD guideline PDF](#).

2.2 Key New Studies

Study / source	Design	Key findings or relevance	GRADE Quality
Implementing the World Health Organization Guidelines for Rheumatic Heart Disease in Highly Endemic Settings: Balancing Evidence with Reality	Implementation-focused article/commentary	Discusses practical challenges in applying WHO RHD guidance in highly endemic settings; highlights feasibility, diagnostic yield, and adaptation issues Global Heart	Very Low
Providencia et al., Diagnostic test accuracy of simplified algorithms for diagnosing acute rheumatic fever: a systematic review , 2025	Systematic review of diagnostic test accuracy	Potentially relevant to WHO recommendations on RF diagnosis and Jones criteria; supplied results did not provide pooled sensitivity/specificity or certainty assessment Global Heart	Low
Birru et al., Severe adverse reactions to benzathine penicillin G in rheumatic heart disease: a systematic review and meta-analysis , 2025	Systematic review and meta-analysis	Relevant to safety of long-term benzathine penicillin G prophylaxis; supplied results did not provide quantitative adverse-event estimates Global Heart	Low
Hirani et al., Acute rheumatic fever , The Lancet, 2025	Narrative/review article	Provides contemporary overview of ARF; useful for surveillance but not identified as a systematic review or RCT in supplied results Global Heart	Very Low
2025 Sydenham’s chorea case report/literature review	Case report and review	Describes an 11-year-old with Sydenham’s chorea treated with antibiotics, NSAIDs, and valproic acid; low-level evidence for diagnosis/management context Scientific Research Publishing	Very Low

Study / source	Design	Key findings or relevance	GRADE Quality
Clinical trial registry updates	Registry search signal	No new RCTs or registry updates were identified in the supplied results for 29 October 2024–29 April 2026	Very Low

2.3 Evidence Gaps and Future Research Needs

1. Acute rheumatic fever diagnosis

- WHO continues to recommend the Jones criteria for RF diagnosis, but implementation literature notes that diagnostic criteria may miss some cases in highly endemic settings [Global Heart](#).
- Future research should evaluate simplified ARF diagnostic algorithms in high-burden primary-care and district-hospital settings.

2. Echocardiographic screening

- WHO gives a strong recommendation with high-certainty evidence for echocardiographic screening among children and adolescents aged **5–19 years** in moderate/high-prevalence settings [NCBI Bookshelf](#).
- Evidence remains very low certainty for antenatal screening, requiring stronger data on outcomes, cost-effectiveness, referral completion, and maternal-fetal benefit.

3. Antibiotic prophylaxis safety and adherence

- Long-term antibiotic prophylaxis is strongly recommended for RF/RHD, but implementation barriers include adherence, access, injection pain, supply chains, and safety concerns.
- The 2025 systematic review/meta-analysis on severe adverse reactions to benzathine penicillin G should be fully appraised in future WHO surveillance [Global Heart](#).

4. Skin and skin-structure infection pathways

- WHO could not make recommendations for or against specific clinical prediction rules or antibiotic treatment of skin infections specifically to prevent RF/RHD [NCBI Bookshelf](#).
- Trials and prospective cohort studies are needed in endemic settings to clarify whether treating GAS skin infection reduces RF/RHD incidence.

5. Youth and lived-experience engagement

- RHD commonly begins in childhood and peaks diagnostically in young adults aged **20–39 years** [WHO RF/RHD guideline](#).
- Future RF/RHD guideline updates should consider applying WHO’s meaningful youth engagement principles, although the existing WHO youth-engagement case study is based on abortion care rather than CVD [WHO publication page](#).

2.4 Updated Recommendations Based on New Evidence

Original WHO recommendation	New evidence impact through 29 April 2026	Updated recommendation
Use Jones criteria for RF diagnosis.	A 2025 diagnostic-accuracy systematic review of simplified ARF algorithms was identified, but supplied results do not provide sufficient accuracy estimates or certainty to change WHO guidance.	No change. Continue Jones criteria while monitoring evidence on simplified algorithms.
Use handheld echocardiography where standard echocardiography is unavailable.	No new evidence in supplied results contradicts this recommendation.	No change. Continue handheld echocardiography as an access-expanding diagnostic tool where standard echo is unavailable.
Screen children/adolescents aged 5–19 years in moderate/high-prevalence settings using standard or handheld echocardiography.	No post-guideline evidence identified that weakens the high-certainty basis of this recommendation.	No change. Continue screening in moderate/high-prevalence settings with referral pathways.
Consider antenatal echocardiographic screening in moderate/high-risk settings.	No new high-certainty evidence found; evidence remains very low certainty.	No change. Keep as conditional; implement where feasible with linkage to maternal cardiac care.
Provide antibiotic prophylaxis to people diagnosed with RF or RHD.	A 2025 systematic review/meta-analysis on severe benzathine penicillin G reactions was identified but quantitative results were not available in supplied summaries.	No change. Continue prophylaxis; strengthen safety protocols, allergy assessment, adverse-event preparedness, and adherence support.
No recommendation for or against antibiotic treatment of skin infections specifically to prevent RF/RHD.	No new RCTs or decisive evidence identified.	No change. Treat skin infections according to local infectious disease standards; do not claim proven RF/RHD prevention benefit until evidence improves.
WHO should monitor new evidence and reassess within 5 years unless earlier evidence triggers an update.	No official WHO update identified through 29 April 2026.	No change. Maintain active evidence surveillance, especially for low- and very-low-certainty areas.

Practical Clinical and Public-Health Implementation Summary

For clinicians

1. Assess CVD risk using a total-risk approach whenever possible.
2. Detect and manage hypertension, diabetes, dyslipidaemia, tobacco use, unhealthy diet, harmful alcohol use, and physical inactivity.
3. In RF/RHD-endemic settings:
 - Use Jones criteria for suspected RF.
 - Use echocardiography, including handheld echocardiography when standard echo is unavailable.
 - Provide long-term antibiotic prophylaxis for diagnosed RF/RHD.
 - Support adherence and safe benzathine penicillin G administration.

For health systems

1. Implement WHO HEARTS primary-care models for CVD prevention and management [WHO HEARTS risk-based CVD management](#).
2. Use standardized hypertension protocols in primary care [WHO HEARTS hypertension protocol tool](#).
3. Prioritize high-risk case-finding over broad untargeted CVD screening where aligned with WHO evidence summaries [WHO HEN report](#).
4. Monitor CVD programme indicators using tools such as SEAHEARTS 2030 where regionally applicable [WHO CVD topic](#).

Final recommendation status as of 29 April 2026

No evidence identified in the supplied materials requires a change to the latest WHO RF/RHD guideline recommendations. The strongest current WHO-based CVD priorities remain:

- Total cardiovascular-risk assessment.
- Primary-care hypertension control.
- Lifestyle and medication-based risk-factor management.
- RF/RHD prevention, diagnosis, screening, and secondary prophylaxis in endemic settings.
- Strengthened implementation through HEARTS and monitoring frameworks.